

- What is the DOM?
- Key Objects:
- DOM vs BOM:
- DOM Node Types
 - Node Properties:
- Element Selection
 - Single Element Selectors:
 - Multiple Element Selectors:
 - Special Selectors:
- DOM Traversal
 - Upward Traversal:
 - Downward Traversal:
 - Sideways Traversal:
- Node Manipulation
 - Creating Nodes:
 - Modifying DOM:
 - Content Manipulation:
- Attribute & Class Manipulation
 - Standard Attributes:
 - Data Attributes:
 - Class Operations:
- Style Manipulation
 - Inline Styles:
 - Computed Styles:
- Event System
 - Event Registration:
 - Event Object:
 - Event Control:
- Complete Event Reference
 - Mouse Events:
 - Keyboard Events:
 - Form Events:
 - Window Events:
- DOM Performance
 - Best Practices:
- Shadow DOM
 - Creating Shadow Roots:

- [Mutation Observer](#)
 - [Watching DOM Changes:](#)

What is the DOM?

- Tree representation of HTML documents
- Programming interface for web documents
- Live data structure that can be modified

Key Objects:

```
document;           // Entry point
document.doctype;   // Document type
document.documentElement; // <html>
document.head;      // <head>
document.body;      // <body>
```

DOM vs BOM:

DOM (Document Object Model)	BOM (Browser Object Model)
document, nodes, elements	window, navigator, history
HTML structure	Browser capabilities

DOM Node Types

```
Node.ELEMENT_NODE (1)
Node.TEXT_NODE (3)
Node.COMMENT_NODE (8)
Node.DOCUMENT_NODE (9)
Node.DOCUMENT_TYPE_NODE (10)
Node.DOCUMENT_FRAGMENT_NODE (11)
```

Node Properties:

```
node.nodeType
node.nodeName
node.nodeValue
node.textContent
node.parentNode
node.childNodes
node.firstChild
node.lastChild
node.previousSibling
node.nextSibling
```

Element Selection

Single Element Selectors:

```
document.getElementById('id');
document.querySelector('css-selector');
```

Multiple Element Selectors:

```
document.getElementsByClassName('class'); // Live HTMLCollection
document.getElementsByTagName('div');      // Live HTMLCollection
document.getElementsByName('name');       // Live NodeList
document.querySelectorAll('selector');    // Static NodeList
```

Special Selectors:

```
document.forms;      // All forms
document.images;     // All images
document.links;      // All links
document.scripts;    // All scripts
```

DOM Traversal

Upward Traversal:

```
element.parentElement;  
element.parentNode;  
element.closest('selector'); // Nearest ancestor
```

Downward Traversal:

```
element.children;           // Element children only  
element.childNodes;        // All node types  
element.firstElementChild;  
element.lastElementChild;
```

Sideways Traversal:

```
element.previousElementSibling;  
element.nextElementSibling;
```

Node Manipulation

Creating Nodes:

```
document.createElement('div');  
document.createTextNode('text');  
document.createDocumentFragment();  
document.createAttribute('attr');  
element.cloneNode(true); // Deep clone
```

Modifying DOM:

```
parent.appendChild(node);
parent.insertBefore(new, reference);
parent.replaceChild(new, old);
parent.removeChild(node);
node.remove(); // Modern alternative
```

Content Manipulation:

```
element.innerHTML = '<b>HTML</b>';
element.textContent = 'Plain text';
element.outerHTML = '<div>Replacement</div>';
```

Attribute & Class Manipulation

Standard Attributes:

```
element.setAttribute('name', 'value');
element.getAttribute('name');
element.removeAttribute('name');
element.hasAttribute('name');
element.attributes; // NamedNodeMap
```

Data Attributes:

```
element.dataset.customName; // data-custom-name
```

Class Operations:

```
element.classList.add('new');  
element.classList.remove('old');  
element.classList.toggle('active');  
element.classList.contains('check');  
element.className = 'class1 class2'; // Legacy
```

Style Manipulation

Inline Styles:

```
element.style.color = 'blue';  
element.style.cssText = 'color: blue; font-size: 16px;';
```

Computed Styles:

```
window.getComputedStyle(element).property;
```

Event System

Event Registration:

```
element.addEventListener('click', handler, {  
  capture: false, // Default (bubbling phase)  
  once: true,     // Auto-remove after trigger  
  passive: true   // For scroll performance  
});
```

Event Object:

```
event.type;  
event.target;  
event.currentTarget;  
event.eventPhase;  
event.timeStamp;  
event.bubbles;  
event.cancelable;
```

Event Control:

```
event.preventDefault(); // Stop default action  
event.stopPropagation(); // Stop bubbling/capturing  
event.stopImmediatePropagation(); // Also stops other handlers
```

Complete Event Reference

Mouse Events:

```
click, dblclick, mousedown, mouseup, mousemove  
mouseover, mouseout, mouseenter, mouseleave  
contextmenu, wheel
```

Keyboard Events:

```
keydown, keyup, keypress
```

Form Events:

```
submit, reset, change, input, invalid  
focus, blur, focusin, focusout
```

Window Events:

```
load, DOMContentLoaded, resize, scroll  
unload, beforeunload, hashchange
```

DOM Performance

Best Practices:

```
const buttons = document.querySelectorAll('.btn');  
  
document.body.addEventListener('click', (e) => {  
  if (e.target.matches('.btn')) {  
    // Handle button click  
  }  
});
```

Shadow DOM

Creating Shadow Roots:

```
const shadow = element.attachShadow({ mode: 'open' });  
shadow.innerHTML = `<style>/* Scoped CSS */</style><slot></slot>`;
```

Mutation Observer

Watching DOM Changes:


```
const observer = new MutationObserver((mutations) => {
  mutations.forEach(mutation => {
    console.log(mutation.type, mutation.target);
  });
});
observer.observe(element, {
  attributes: true,
  childList: true,
  subtree: true,
  characterData: true
});
```